

Experiment 'p_4_3_c_replace_vowels' Results

June 3, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 2

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 2

Problem Specification

Problem name: p_4_3_c_replace_vowels

Natural language statement: Write a method that takes a string and returns the string, with all vowels replaced by an underscore.

Method signature: p_4_3_c_replace_vowels(s: string) returns (result: string)

Ensures

- `|result| == |s|`
- forall `i :: 0 <= i < |s| ==> (if s[i] in ['a','e','i','o','u','A','E','I','O','U'] then result[i] == '_' else result[i] == s[i])`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes a string and returns the string, with all vowels replaced by an
→ underscore.

The signature should be:

p_4_3_c_replace_vowels(s: string) returns (result: string)

The method should respect the following contract:

ensures |result| == |s|, ensures forall i :: 0 <= i < |s| ==> (if s[i] in ['a','e','i','o'
→ ','u','A','E','I','O','U'] then result[i] == '_' else result[i] == s[i])

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmenXuJ8BxUJ4EJEvSUAETMjjoUcl

Model: gpt-5.5-2026-04-23

Created at: 1780488943

Finish reason: stop

Usage: 382 tokens in, and 1431 tokens out

```
{
  result := seq(|s|, i => if s[i] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U']
    → then '_' else s[i]);
}
```

Program → GenAI

When using dafny verify, the below error is emitted and verify fails:

```
problem_data/p_4_3_c_replace_vowels/program.dfy(5,30): Error: index out of range
|
5 | result := seq(|s|, i => if s[i] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U']
    → then '_' else s[i]);
| ^
```

Dafny program verifier finished with 1 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
→ Dafny body, including the curly braces that surround it. Do not show the signature
→ nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmeoH1djHlDP0Feu753S1wdlPzInl

Model: gpt-5.5-2026-04-23

Created at: 1780488989

Finish reason: stop

Usage: 623 tokens in, and 612 tokens out

```
{
  result := seq(|s|, i =>
    if 0 <= i < |s| && s[i] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U'] then '_'
    → else
    if 0 <= i < |s| then s[i] else '_');
}
```

Final Program

```
method p_4_3_c_replace_vowels(s: string) returns (result: string)
  ensures |result| == |s|
  ensures forall i :: 0 <= i < |s| ==> (if s[i] in ['a','e','i','o','u','A','E','I',
    → ',','O','U'] then result[i] == '_' else result[i] == s[i])
{
  result := seq(|s|, i =>
    if 0 <= i < |s| && s[i] in ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U'] then '_'
    → else
    if 0 <= i < |s| then s[i] else '_');
}
```

Total Token Usage

Input tokens: 1005

Output tokens: 2043

Reasoning tokens: 1858

Sum of 'total tokens': 3048

Experiment Timings

Overall Experiment started at 1780488943701, ended at 1780489008861, lasting 65160ms (65.16 seconds)

Iteration #1 started at 1780488943701, ended at 1780488989275, lasting 45574ms (45.57 seconds)

Iteration #2 started at 1780488989275, ended at 1780489008861, lasting 19586ms (19.59 seconds)